

A Retained Mixed Gravity–Maxwell Bracket under Cyclic Homological Reduction through ℓ_3 Exact Transfer and Causal Cartan Compatibility

GPT-5.6.sol*

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Abstract

Does the mixed gravity–Maxwell interaction of a fourth-order clock background survive homological reduction to a small complex? We transfer the classical gravity–Maxwell BV interaction on a compact positive Berger-clock background from a complete 64-row presentation to a typed cyclic 36-row retained complex, carrying the exact odd pairing and the unary causal Green data. For this explicitly specified cyclic strong deformation retract the retained mixed ternary operation ℓ_3 is nonzero, so the retained mixed gravity–Maxwell bracket survives the reduction through ℓ_3 .

Nonvanishing is witnessed by two explicit zero-derivative evaluations, with exact coefficients $+1$ and -1 in the gravity-equation and Maxwell-equation sectors, rather than inferred from a sparse-coefficient count. The homological exchange channel is computed rather than omitted: its only nonempty channel is annihilated by the certified retained projection, so the retained exchange contribution vanishes for a structural reason while the direct q_3 contact survives. Standard cyclic homotopy transfer supplies the abstract L_∞ and cyclic identities, and an independent sparse implementation audit replays them on all 36 rows, closing cyclicity of the full retained BV tensor. The separately certified causal theorem gives compatible advanced and retarded unary homotopies and two-sided-causal-hull Cartan primitives for the background-fixing helical generator K_{Berger} . It is a parallel compatibility theorem: ℓ_3 is transferred with the support-local algebraic homotopy S , not by integrating out fields with a causal propagator.

The result is classical and carries separate LOCAL-ALGEBRAIC and LORENTZIAN-CAUSAL dependencies. Its limits are sharp and we state them. It does not prove a nonzero operation on cohomology, invariance under changing the cyclic retract, or nonremovability under every unrestricted cyclic L_∞ equivalence. The first deformation-filtration screen is exact and negative: after setting every PBW derivative word to zero, the complete mixed $g^2 A^2$ quartic sector is the image of admissible constant-field cyclic redefinitions, so the displayed zero-derivative evaluations witness representative nonvanishing but cannot witness deformation-class nontriviality. The zero-PBW full-BV page is likewise exact after adjoining three certified Maxwell covariant-ghost shears, and the complete degree-zero physical action through summed input-jet order two has an explicit primitive; neither statement alone is a full-BV positive-jet theorem. A previously reported 22-row order-two obstruction does not survive the declared derivative-aware cyclic complex and its promotion to total order two is withdrawn here, with the earlier replay remaining valid only on the zero- and first-jet columns it actually tested. The complete bounded cyclic deformation class is therefore open: no replacement cokernel witness and no complete full-BV trivializing cochain is claimed. Residual mixing, Hadamard products, QME restoration, and every

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quantum or particle interpretation remain open. Finally, a separate exact audit obstructs the canonical local same-bundle branch projector on the 36-row carrier; the resulting spatial-STF2 graph prolongation and its cyclic dual are constructed as an exact local cyclic carrier whose principal wave module is a nonsplit Einstein/extra-Weyl filtration, so the next branch gate is the exact subprincipal extension rather than a principal direct-sum projector.

AI authorship and computational accountability

The named OpenAI model developed the mathematical synthesis, verification code, claim auditing, and manuscript. Asger Alstrup Palm commissioned and coordinated the programme and is the corresponding human contact. Every numerical count in the theorem is an exact sparse-coefficient count over $\mathbb{Q}(\sqrt{10})$ and is tied to a deterministic certificate. No floating point calculation enters an identity, rank, cyclicity test, or mutation witness. Formal journal submission remains conditional on expert scope and presentation review.

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1 Question and answer

The free BV complex identifies which classes survive gauge reduction, but it does not determine whether nonlinear gravity–matter vertices survive the same reduction. A transferred interaction can disappear in three distinct ways: its direct contact term can project to zero; homological exchange terms can cancel it; or cyclicity and the L_∞ identities can fail after compression. Conversely, a nonzero tensor on a gauge-fixed carrier need not descend to any physical branch until its residual projection is computed.

This paper answers the algebraic question through ternary order for one explicit gravity–Maxwell extension and one frozen cyclic contraction of the positive Berger-clock complex. Write

$$\boxed{\text{The retained mixed gravity–Maxwell bracket survives this cyclic homological reduction through } \ell_3.} \quad (1)$$

Here “survives” means that the explicitly transferred mixed ternary bracket is nonzero in the displayed retained coordinates and obeys the transferred arity-three identity. “Cyclic” means that the contraction is typed cyclic and that the complete lowered BV quartic form has been independently replayed under cyclic transposition. Causal Green and Cartan compatibility are proved separately in Section 6; they are not part of the definition of the algebraic transfer.

Nonvanishing in Theorem 4.1 is deliberately representative-level. By itself it does not prove

$$[\ell_3^{\text{mixed}}] \neq 0 \quad (2)$$

in a cyclic coderivation or Chevalley–Eilenberg deformation complex. Nor have we proved a nonzero induced operation on $H(\mathcal{R}_{36}, \ell_1)$, survival under all cyclic L_∞ equivalences or further minimal transfer, or an obstruction to removing the selected component under every cyclic change of retained coordinates. Section 8 gives the exact current disposition. The physical action is removable through input order two, while an admissible second-jet cyclic column invalidates the only proposed full-BV dual witness. This reopens rather than decides Equation (2).

The theorem stops one rung before physical branch interpretation. A subsequent exact audit now obstructs the canonical support-local same-bundle Einstein-like/extra-Weyl projector on the 36-row carrier. Residual descent therefore requires an enlarged or differently organized carrier. The natural rank-46 STF2 graph carrier has now passed its exact unary cyclic-SDR gate. Its principal direct-sum anchor is obstructed, while the lower-order Berger splitting and nonlinear lift remain open, so a branch mixing table is still unauthorized. Deformation/vertex classes belong to a separate complex and are not used in the theorem below.

2 Complexes, conventions, and claim boundary

Let $(\mathcal{C}_{64}, q_1, \omega_{64})$ be the complete typed classical gravity–clock–Maxwell BV complex at the frozen rational Berger fixture. Its degree multiplicities are

$$6, 26, 26, 6 \quad \text{in degrees } -1, 0, 1, 2. \quad (3)$$

Let $(\mathcal{R}_{36}, \ell_1, \omega_{36})$ denote the retained typed carrier, with degree multiplicities

$$4, 14, 14, 4. \quad (4)$$

The exact contraction is

$$(\mathcal{C}_{64}, q_1) \begin{matrix} \xrightarrow{\pi} \\ \xleftarrow{\iota} \end{matrix} (\mathcal{R}_{36}, \ell_1), \quad \pi\iota = \text{id}, \quad q_1 S + S q_1 = \text{id} - \iota\pi, \quad (5)$$

with the exact side conditions

$$S^2 = 0, \quad S\iota = 0, \quad \pi S = 0, \quad (6)$$

and typed cyclic adjoint relations fixed by the carrier certificate. The coefficients lie in $\mathbb{Q}(\sqrt{10})$. The Maxwell field–equation pairing has absolute weight two, while the gravity weight is one. The

signed odd-pairing ledger contains ten gravity entries of sign -1 and four Maxwell entries of sign $+2$ on the degree-zero field–equation pairs.

For completeness, let $\dagger$ denote the formal PBW adjoint, including integration by parts, and let Ω_{64} and Ω_{36} be the signed typed pairing matrices. All suspension and Koszul signs are already included in these matrices. The exact convention-specific cyclic SDR relations are

$$\iota^\dagger \Omega_{64} = \Omega_{36} \pi, \quad \iota^\dagger \Omega_{64} \iota = \Omega_{36}, \quad S^\dagger \Omega_{64} + \Omega_{64} S = 0. \quad (7)$$

Equivalently, for homogeneous retained x and full-carrier y ,

$$\omega_{64}(\iota x, y) = \omega_{36}(x, \pi y), \quad \omega_{64}(Sx, y) = -\omega_{64}(x, Sy). \quad (8)$$

The last displayed sign is therefore the suspended-coordinate sign actually verified by the certificate, not an unsuspended sign imported from another convention.

Table 1 gives the carrier map needed to interpret the row indices. The retained gravity representatives are dressed combinations, so the table states their image by bundle type rather than pretending that the gravity part of ι is a coordinate inclusion. The Maxwell summand is retained by the identity.

Table 1: Human-readable carrier map. A zero retained image means that the displayed full block belongs to the contractible complement.

Full rows	Degree	Full carrier content	Retained image
0–4	-1	three spatial diffeomorphism ghosts, clock ghost τ , Weyl ghost σ	spatial ghosts 0–2
5–16	0	ten $\widehat{h}_{\mu\nu}$ fields, R , Θ	dressed $\widehat{h}$ rows 3–12
17–26	0	nonminimal multipliers and antighost-duals	0
27–37	1	ten $\widehat{h}_{\mu\nu}^*$ equations and R^*	dressed $\widehat{h}^*$ rows 13–22
38	1	Θ^* , the clock/Stueckelberg equation dual	$\mathbf{0}$ under π
39–48	1	nonminimal dual and antighost rows	0
49–53	2	three spatial ghost antifields, τ^* , σ^*	spatial ghost-antifields 23–25
54	-1	Maxwell ghost c_M	c_M (row 26)
55–58	0	Maxwell fields $A_0, \dots, A_3$	rows 27–30
59–62	1	Maxwell equations $A_0^+, \dots, A_3^+$	rows 31–34
63	2	Maxwell ghost antifield c_M^+	row 35

The row-38 support statement is therefore geometric. Rows 3, 16, 38, 52 of the full carrier are $\tau, \Theta, \Theta^*, \tau^*$, and the certified blocks include

$$\mathrm{pr}_\Theta q_1(\tau) = \Theta, \quad q_1(\Theta^*) = -\tau^*, \quad S(\Theta) = \tau, \quad S(\tau^*) = -\Theta^*. \quad (9)$$

Moreover

$$\pi|_{\langle \Theta^*, \tau^* \rangle} = 0, \quad \mathrm{pr}_{\langle \Theta^*, \tau^* \rangle} \iota = 0, \quad \omega_{64}(\Theta, \Theta^*) = 1, \quad \omega_{64}(\tau, \tau^*) = 1. \quad (10)$$

Thus row 38 is one half of the cyclic dual contractible clock doublet, not an unnamed discarded equation. At block level, ι , π , and S have respectively 86, 86, and 19 nonzero row-pair blocks; the vanishing in (10) is checked against those complete support ledgers.

The based classical BV vector field is known through cubic Taylor order,

$$q = q_1 + q_2 + q_3 + O(4), \quad (11)$$

in the repository’s suspended graded-symmetric factorial convention. In this convention every q_n and ℓ_n has cohomological degree $+1$. The full tensors are action-derived, support-local, typed cyclic, and equivariant under the background-fixing helical generator

$$K_{\text{Berger}} = D - \omega R. \quad (12)$$

Raw cylinder translation D is affine at the rotating background and is not substituted for K_{Berger} .
The result combines two dependency layers:

Layer	Certified content	Explicit limitation
LOCAL-ALGEBRAIC	exact q_2, q_3, ℓ_2, ℓ_3 , identities and cyclicity	no residual branch interpretation
LORENTZIAN-CAUSAL	unary Green homotopies and K -Cartan support through arity three	no Hadamard products or QME

Neither layer is silently promoted to RESIDUAL-TRANSFERRED or LORENTZIAN-CERTIFIED quantum status.

3 Ternary homological transfer

Proposition 3.1 (Formal cyclic transfer through arity three). *Let a cyclic L_∞ algebra be given through q_3 , and let (π, ι, S) be a cyclic SDR satisfying (5)–(6) and (7). The rooted-tree formulas (14)–(17) define cyclic operations through ternary order and imply*

$$[\ell_1, \ell_3] + \frac{1}{2}[\ell_2, \ell_2] = 0. \quad (13)$$

This is the standard cyclic homotopy-transfer theorem [3, 4]. Its application is the mathematical argument for the identities. The sparse calculations below are a separate implementation certificate: they test that the declared signs, PBW normalization, carrier maps, and typed pairing actually realize the formal theorem.

The first two transferred Taylor operations are

$$\ell_1 = \pi q_1 \iota, \quad (14)$$

$$\ell_2 = \pi q_2(\iota, \iota). \quad (15)$$

Define the second Taylor component of the inclusion by

$$I_2 = -Sq_2(\iota, \iota). \quad (16)$$

Then the ternary transfer formula is

$$\ell_3 = \pi q_3(\iota, \iota, \iota) + \sum_{\sigma \in \text{Sh}(2,1)} \epsilon(\sigma) \pi q_2(I_2(\iota_{\sigma(1)}, \iota_{\sigma(2)}), \iota_{\sigma(3)}). \quad (17)$$

The first term is the contact contribution. The second line is the exchange contribution $q_2 Sq_2$.

Lemma 3.2 (Exact mixed exchange support for the frozen SDR). *For the mixed gravity–Maxwell ternary operation, the gravity-outer/mixed-inner exchange channel contains 144 outer/inner co-efficient pairs. The three graded unshuffles produce 324 signed contributions and 342 canonical full-complex PBW coefficients. Every one has full output row $38 = \Theta^*$. The retained projection has no support on that row, so this channel vanishes after projection. The mixed-outer/gravity-inner and mixed-outer/mixed-inner channels have no outer/inner pairs.*

Proof. The exact consumer reconstructs the 96 gravity and 12 mixed coefficients of I_2 , including their full-row supports, before composing them with q_2 . Outer Leibniz expansion can split one signed unshuffle contribution among several derivative distributions before canonical collection; this is why 324 signed contributions yield 342 PBW coefficients. Direct application of π annihilates the 342 row-38 coefficients. As a negative control, adding the single mutant entry

$$\pi_{\text{mut}}(\text{retained row 0, full row 38}) = 1 \quad (18)$$

exposes all 342 coefficients. Thus the zero is caused by the certified projection and is not an empty table hard-coded by the consumer. $\square$

Corollary 3.3 (Contact form for the frozen retained coordinates). *The retained mixed exchange contribution is zero and*

$$\ell_3^{\text{mixed}} = \pi q_3^{\text{mixed}}(\iota, \iota, \iota). \quad (19)$$

This equality is a computed property of the declared contraction, not a general truncation of the homotopy-transfer formula. In particular, a different cyclic homotopy or cyclic L_∞ coordinate change may redistribute a representative between contact and exchange trees.

4 Frozen-SDR nonvanishing and its audit

For a retained basis vector u , write $[\ell_3(x, y, z)]_u^{(0)}$ for the exact coefficient of u with the zero PBW derivative word in every input slot.

Theorem 4.1 (Nonzero retained mixed bracket for the frozen cyclic SDR). *On the frozen positive Berger-clock gravity–Maxwell BV complex, the typed cyclic contraction (5) transfers the action-derived Taylor data through ternary order. Its mixed ternary representative is nonzero, with the two exact evaluations*

$$[\ell_3(\widehat{h}_{13}, A_1, A_3)]_{\widehat{h}_{00}^*}^{(0)} = +1, \quad (20)$$

$$[\ell_3(\widehat{h}_{01}, \widehat{h}_{03}, A_3)]_{A_1^+}^{(0)} = -1. \quad (21)$$

The first is a gravity-equation output with one gravity and two Maxwell inputs; the second is a Maxwell-equation output with two gravity and one Maxwell input. Pairing the outputs with $\widehat{h}_{00}$ and A_1 , respectively, makes both witnesses explicit coefficients in the lowered g, g, A, A quartic sector. All mixed gravity–Maxwell exchange channels contributing to this declared ℓ_3^{mixed} vanish for the frozen SDR by Lemma 3.2. The operations obey the arity-three L_∞ identity (13) and cyclicity under the retained pairing.

Proof. The two displayed coefficients are read from the exact independently replayed retained tensor; their output rows are 13 and 32, and their input rows are (9, 28, 30) and (4, 6, 30). All inputs have degree zero, both outputs have degree one, and the pairing partners are retained rows 3 and 28. Hence the evaluations are typed and nonzero over $\mathbb{Q}(\sqrt{10})$ without reference to a total coefficient count. Lemma 3.2 proves the scoped exchange statement. Proposition 3.1 and the certified cyclic SDR give the abstract L_∞ and cyclic conclusions. $\square$

The gravity labels in Equations (20) and (21) denote dressed retained coordinates. Their exact full-carrier inclusion columns, including every nonminimal gauge-fixing tail, are printed in Table 2. The Maxwell columns are identity columns. This prevents the witness notation from being mistaken for a raw-coordinate inclusion.

Proposition 4.2 (Constant-field cyclic-redefinition screen). *Let ev_0 set every PBW derivative word to zero after lowering the retained operations with the typed odd pairing. On the complete mixed quartic space*

$$\text{Sym}^2(G^*) \otimes \text{Sym}^2(A^*), \quad \dim = 550, \quad (22)$$

the matter-parity-preserving, zero-jet BV-canonical cotangent-lift ansatz has 810 coefficients in F_2 and 1,880 in F_3 . Its infinitesimal coboundary map

$$\delta S_4 = \sum_i (\partial_i S_2) F_3^i + \sum_i (\partial_i S_3) F_2^i \quad (23)$$

has exact rank 550 over $\mathbb{Q}(\sqrt{10})$. The landed constant-field mixed quartic has 63 nonzero coordinates and is reconstructed exactly by an exported 51-coefficient primitive supported entirely in F_3 .

Proof. The exact solver aggregates the frozen factorial-convention tensors into commutative lowered action polynomials, differentiates S_2 and S_3 , and enumerates all 2,690 matter-parity-preserving base-field maps. The resulting $550 \times 2,690$ matrix has 5,105 nonzero entries. A deterministic support matching selects a 550-column block whose rank is then checked exactly; no structural or floating-point rank is used. The independent verifier rebuilds the action polynomials and coboundary matrix from the pinned ℓ_1, ℓ_2, ℓ_3 payloads, rechecks the rank basis, and multiplies the exported primitive into all 550 target coordinates. A one-coefficient primitive mutation fails the reconstruction. $\square$

Proposition 4.2 is a G0 filtration result, not a trivialization of the full retained operation. In particular, derivative terms in F_2, F_3 , PBW integration-by-parts relations, and cancellation of higher-order terms against the fourth-order unary operator have not yet been solved, and the 288 ghost/antifield completion coefficients have not been independently matched by this primitive.

Proposition 4.3 (First positive-jet cyclic-redefinition screen). *After exact first-order integration by parts, the mixed physical quartic has 1,330 first-jet coordinates per PBW axis. Let A denote the zero-jet map of Proposition 4.2, B_a the first-jet effect of its complete 2,690-coefficient zero-jet ansatz, and C the map from first-jet redefinitions. The coupled equations*

$$Ax_0 = t_0, \quad B_ax_0 + Cy_a = t_{1,a}, \quad a = 0, 1, 2, 3, \quad (24)$$

have an exact solution over $\mathbb{Q}(\sqrt{10})$. More precisely, C has shape $1,330 \times 6,560$, rank 1,327, and cokernel dimension three. The cokernel-reduced Schur system has shape $562 \times 2,690$, rank 557, and compatible target. An exported primitive has 51 zero-jet coefficients and respectively 43, 94, 95, 108 first-jet coefficients on the four PBW axes.

Proof. The solver lowers the pinned ℓ_1, ℓ_2, ℓ_3 tensors to local action densities and quotients total first derivatives with exact field-multiplicity weights. It projects Equation (24) to the three exact cokernel rows of C on each axis before solving for x_0 ; hence it retains the entire affine zero-jet solution space rather than extending the earlier chosen primitive by fiat. Exact rank receipts and direct multiplication of the exported x_0, y_a reconstruct the 550 zero-jet coordinates and all four 1,330-coordinate first-jet blocks. A primitive-coefficient mutation fails the replay. $\square$

This remains a physical-action filtration theorem. The later certified Maxwell covariant-ghost shear supplies the smallest missing zero-PBW full-BV carrier: an exact 67-coefficient F_2 primitive reconstructs all 186 canonical mixed zero-word BV coefficients. Separately, the complete degree-zero physical-action problem through summed input-jet order two has an exact 4,276-coefficient homogeneous correction. It replays all 550 zero-page equations, four 1,330-row first-page blocks,

and 10,043 nonzero order-two Euler coordinates. This latter primitive is physical-base only; its derivative-aware positive-jet ghost/antifield lift is precisely the open full-BV gate disposed in Section 8.

Corollary 4.4 (Exact sparse implementation audit). *The independent PBW consumer realizes Theorem 4.1 with the following basis-level audit data.*

1. *The retained coupled binary operation ℓ_2 contains 1,474 canonical coefficients.*
2. *The retained mixed ternary operation ℓ_3 contains 25,950 canonical coefficients on 18 nonzero output rows.*
3. *All mixed gravity–Maxwell exchange channels contributing to the declared mixed ℓ_3 vanish for the support reason in Lemma 3.2.*
4. *Direct coefficient comparison of the retained arity-three identity closes on all 36 rows:*

$$[\ell_1, \ell_3] + \frac{1}{2}[\ell_2, \ell_2] = 0. \quad (25)$$

5. *The mixed bracket has 7,614 gravity-output coefficients with two Maxwell inputs and 18,336 Maxwell-output coefficients with one Maxwell input.*

A localized one-coefficient mutation of ℓ_3 creates two exact arity-three defects.

The audit is obtained as follows. Equations (15)–(17) are evaluated in exact sparse PBW normal form over $\mathbb{Q}(\sqrt{10})$. The independent consumer parses all 59,598 coefficients of the full mixed q_3 payload and reproduces all 25,950 retained contact coefficients with no missing, extra, or changed terms. Lemma 3.2 evaluates every possible mixed exchange channel. Direct comparison of (25) gives zero defects on all rows. The input-sector and output-sector ledgers give the two sector counts. Finally, the mutation changes one retained coefficient and produces two normalized defects, preventing acceptance by a consumer that merely assumes the expected answer.

The numbers in Corollary 4.4 are reproducibility metadata, not invariant content: they depend on the retained basis, row ordering, PBW and integration-by-parts normalization, and the chosen SDR. The invariantly stronger questions in Equation (2) remain open. In particular, Theorem 4.1 is weaker than a residual particle vertex because $\mathcal{R}_{36}$ is retained, not minimal or decomposed into its dynamical branches.

5 Independent cyclic implementation audits

Lowering the output of ℓ_3 with the odd pairing gives a four-linear functional. For degree-zero retained inputs, cyclic transposition of the first input requires formal adjunction of every PBW derivative through the invariant volume form. If w is a derivative word, the exact formal adjoint distributes its letters over the other three slots with sign $(-1)^{|w|}$ and the corresponding multinomial multiplicities.

Proposition 5.1 (Independent degree-zero lowered cyclicity). *Every degree-zero coefficient of the lowered retained ℓ_3 satisfies exact cyclic transposition. There are*

$$25,662 = 7,506_{\text{gravity output}} + 18,156_{\text{Maxwell output}} \quad (26)$$

such coefficients, and the defect count is zero on every row. Replacing the Maxwell field–equation weight two by weight one produces 17,108 defects on 14 rows.

Proof. The verifier constructs both sides independently from the retained tensor, the typed pairing, and exact PBW integration by parts. The signed pairing orientations are retained as a separate convention ledger; only their absolute component multiplicities enter the degree-zero field-equation transpose. Equality holds coefficientwise. The weight mutation changes no coefficient of ℓ_3 itself and therefore tests the lowering convention rather than the imported interaction. $\square$

The remaining

$$25,950 - 25,662 = 288 \quad (27)$$

coefficients are the ghost/antifield completion. Their retained output rows are

$$23, 24, 25, 26, 32, 33, 34.$$

If x is the transposed first input and u is the coordinate paired with the output, the suspended-Darboux sign is

$$(-1)^{|x||u|+\epsilon_2(x)+\epsilon_2(u)}, \quad \epsilon_2(v) = \begin{cases} 1, & \deg v = 2, \\ 0, & \text{otherwise.} \end{cases} \quad (28)$$

Here the degree-two rows are precisely the retained ghost-antifield coordinates.

Proposition 5.2 (Independent full-BV quartic cyclicity). *The complete 25,950-coefficient lowered retained ℓ_3 satisfies exact cyclic transposition. Within the 288-coefficient completion, 120 coefficients carry the positive transpose sign and 168 carry the negative sign. The exact defect count is zero. Omitting the degree-two polarization term exposes 132 defects on all seven completion output rows.*

This proposition is a full-BV statement about the pinned classical retained tensor. It does not imply full BV local cohomology, a quantum master equation, or a Lorentzian quantum theory.

6 Causal compatibility through ternary order

The causal statement is not inferred from finite PBW tables. It is supplied by the independently certified unary Green chain contractions and the coupled cyclic Cartan theorem, extending the Berger-clock construction of the companion paper [8]. If

$$\Lambda_{36,\pm} = \Lambda_{26,\pm}^{\text{grav+clock}} \oplus \Lambda_{M,\pm}, \quad \Lambda_{M,\pm} = W_M G_{P_M,\pm}, \quad (29)$$

then the certified complete-carrier lift is

$$\Lambda_{64,\pm} = S + \iota \Lambda_{36,\pm} \pi, \quad q_1 \Lambda_{64,\pm} + \Lambda_{64,\pm} q_1 = \text{id}_{64}. \quad (30)$$

The side conditions imply the explicit retained bridge

$$\pi \Lambda_{64,\pm} \iota = \Lambda_{36,\pm}. \quad (31)$$

The advanced and retarded homotopies have the corresponding one-sided causal supports and formal-adjoint reversal. Equivariance is not inferred from the background label: the exact carrier ledgers verify

$$K_{64} \iota = \iota K_{36}, \quad \pi K_{64} = K_{36} \pi, \quad K_{64} S = S K_{64}. \quad (32)$$

Let $\iota_K^{(1)}$ be the unary Cartan primitive. The higher sources are

$$A_K^{(2)} = [q_2, \iota_K^{(1)}] - L_K^{(2)}, \quad (33)$$

$$A_K^{(3)} = [q_3, \iota_K^{(1)}] + [q_2, \iota_K^{(2)}] - L_K^{(3)}. \quad (34)$$

At the frozen background $L_K^{(2)} = L_K^{(3)} = 0$. The L_∞ identities, graded Jacobi identity, and K -equivariance give

$$[q_1, A_K^{(2)}] = [q_1, A_K^{(3)}] = 0. \quad (35)$$

Raw causal primitives are cyclicized with the exact Reynolds projectors

$$\text{Cyc}_3 = \frac{1 + \tau + \tau^2}{3}, \quad \text{Cyc}_4 = \frac{1 + \tau + \tau^2 + \tau^3}{4}. \quad (36)$$

Theorem 6.1 (Cyclic causal Cartan compatibility). *The coupled 64-row complex and its typed 36-row retained contraction obey the K_{Berger} Cartan identities through arity three. The higher cyclic primitives satisfy*

$$\text{supp}(\iota_K^{(n)}(f_1, \dots, f_n)) \subseteq J\left(\bigcup_i \text{supp } f_i\right), \quad n \leq 3. \quad (37)$$

The Taylor operations are support-local and no spatial inverse or mode projector enters the construction.

The higher cyclic primitives use the two-sided causal hull. They are not claimed to be separately retarded or advanced; the unary homotopies retain that stronger one-sided property. Equations (30)–(32) make the bridge between the complete and retained unary causal data explicit. They do not turn the support-local algebraic homotopy S used in (17) into a causal propagator. Thus Theorem 6.1 is a compatibility result parallel to Theorem 4.1, not a construction of ℓ_3 by causal integration. It does not construct interacting retarded products, a Hadamard state, or a Lorentzian quantum master equation.

7 Residual descent and the projector obstruction

The next calculation determines what the surviving tensor means physically. On any admissible enlarged or alternative carrier, the dynamical sector must contain

$$\mathcal{R}_{\text{dyn}} = \mathcal{R}_{\text{Einstein}} \oplus \mathcal{R}_{\text{extra Weyl}} \oplus \mathcal{R}_{\text{Maxwell}}, \quad (38)$$

together with exact inclusion, projection, pairing, parity, real structure, and K_{Berger} weights. The mixing entries are then

$$M^a_{bcd} = \pi_{\text{dyn}}^a \ell_3(\iota_b^{\text{dyn}}, \iota_c^{\text{dyn}}, \iota_d^{\text{dyn}}). \quad (39)$$

Every claimed zero must carry an exact support or cancellation witness, and graded symmetry and cyclicity must be replayed in branch coordinates.

The canonical attempt on $\mathcal{R}_{36}$ has received a binary negative verdict. With the Einstein image fixed by the certified rough tensor-wave branch, the exact endpoint

$$A_{10} = \square_2^2 + V_2, \quad \text{ord}(V_2) \leq 2, \quad (40)$$

has 92 nondivisible degree-two remainder entries. A normalized first witness is

$$\frac{71p_1^2 + 71p_2^2 + 9p_3^2}{80} \mod (-p_0^2 + p_1^2 + p_2^2 + p_3^2), \quad (41)$$

and the normalized functional $(80/71)[p_1^2]$ evaluates it to one. Hence no canonical finite-order support-local same-bundle complementary Einstein-like/extra-Weyl projector of the declared type exists on $\mathcal{R}_{36}$.

This is a scoped obstruction, not a global no-go. Mixed-bundle mapping cylinders, filtered carriers, higher-rank local carriers, and explicitly REDUCED-MODE nonlocal splittings remain open. Exact symbol counting requires at least four additional BV rows; the smallest natural support-local candidate adds a spatial STF2 prolongation and its cyclic dual, giving rank 46.

That candidate has now been constructed. Let T_{STF} extract

$$h_{12}, \quad h_{13}, \quad h_{23}, \quad h_{11} - h_{22}, \quad h_{11} + h_{22} - 2h_{33}, \quad (42)$$

and set $F = T_{\text{STF}}\square_2$. Introducing $Y = Fh$ and its five cyclic-dual rows gives degree ranks $(4, 19, 19, 4)$. The metric Hessian block is

$$\begin{pmatrix} A_{10} + F^\dagger F & -F^\dagger \\ -F & I_5 \end{pmatrix}, \quad (43)$$

whose Schur complement is exactly A_{10} .

Proposition 7.1 (Exact rank-46 STF2 graph carrier). *The finite-order cotangent shear defining Equation (43) and its inverse produce a cyclic strong deformation retract*

$$(\mathcal{R}_{46}, q_1^{46}, \omega_{46}) \xrightleftharpoons[\iota_{36}^{46}]{\pi_{46}^{36}} (\mathcal{R}_{36}, \ell_1, \omega_{36}), \quad S_{46}, \quad (44)$$

with

$$\begin{aligned} (q_1^{46})^2 &= 0, & \pi_{46}^{36} \iota_{36}^{46} &= I_{36}, & q_1^{46} S_{46} + S_{46} q_1^{46} &= I_{46} - \iota_{36}^{46} \pi_{46}^{36}, \\ S_{46}^2 &= 0, & S_{46} \iota_{36}^{46} &= 0, & \pi_{46}^{36} S_{46} &= 0. \end{aligned}$$

Both chain maps, the induced pairing, unary cyclicity, and homotopy cyclicity hold exactly over $\mathbb{Q}(\sqrt{10})$. The added STF2 complement is contractible, so this construction changes neither the retained cohomology nor, by itself, the physical branch content.

The graph extension does not, however, split the repeated physical wave layer at principal order. Put

$$A = \mathbb{Q}(\sqrt{10})[\epsilon]/(\epsilon^2), \quad \epsilon = -p_0^2 + p_1^2 + p_2^2 + p_3^2. \quad (45)$$

For each physical helicity, the Einstein-like layer is the submodule $E = \epsilon A$ and the generalized extra-Weyl layer is the quotient A/E .

Proposition 7.2 (Normalized principal branch-anchor obstruction). *The exact sequence*

$$0 \longrightarrow \epsilon A \longrightarrow A \longrightarrow A/(\epsilon) \longrightarrow 0 \quad (46)$$

does not split A -linearly. Indeed, an attempted section has $s(1) = 1 + b\epsilon$, whereas A -linearity requires $\epsilon s(1) = 0$ and exact multiplication gives

$$\epsilon s(1) = \epsilon. \quad (47)$$

The functional extracting the coefficient of ϵ evaluates this defect to one and annihilates every correction $b\epsilon$. Equivalently, an endomorphism $a + b\epsilon$ is idempotent only for $(a, b) = (0, 0)$ or $(1, 0)$. The real helicity multiplicity is two and the witness has K_{Berger} -weight zero.

This obstruction excludes a nonzero principal direct-sum anchor and prevents the zero, identity, or contractible-auxiliary projector from being promoted as the physical branch split. It is not a no-go for the full rank-46 projector: lower-order Berger terms may deform the repeated leading factor and split it.

The physical quotient needed for that next calculation is now exact. On the normalized projective null cone, the tensorial transverse–traceless idempotent defines a rank-two projective polarization module inside the full six-dimensional degree-zero Berger null-symbol cohomology. Its canonical transpose defines the paired degree-one module, and the odd BV pairing descends nondegenerately. At $\zeta = (1, 1, 0, 0)$ its normalized basis is $\hat{h}_{22} - \hat{h}_{33}, \hat{h}_{23}$ and agrees with the independently certified linearized-Weyl quotient. Thus the polarization rank is two, while tensoring with $\mathbb{Q}(\sqrt{10})[\epsilon]/(\epsilon^2)$ gives a rank-four generalized wave module. No global two-column helicity frame is chosen: the invariant object is the projective module defined by the idempotent.

The filtered order-two extension class defined by V_2 has now been tested by the full symbol-complex descent equation. At $\zeta = (1, 1, 0, 0)$, arbitrary principal gauge changes of the field representative, all principal Hessian boundaries, and both physical equation representatives are included. A normalized left-null witness evaluates on the two physical columns as

$$(1, 0). \quad (48)$$

Thus the plus representative $\hat{h}_{22} - \hat{h}_{33}$ has a nonzero quotient class for every allowed correction. The cross representative $\hat{h}_{23}$ does lift, with physical equation coefficient $71/40$. This is an invariant filtered obstruction, not a diagonalization of the raw 10×10 matrix or the bare compression $\Pi_{\text{TT}} V_2 \Pi_{\text{TT}}$.

The declared rank-46 contractible graph carrier therefore does not supply a closed physical filtered subcomplex or an Einstein-like/extra-Weyl projector with this principal anchor. It also does not supply a q_2, q_3 lift, K_{Berger} equivariance, or causal support on $\mathcal{R}_{46}$. Equation (39) therefore remains a target formula rather than an authorized rank-46 calculation.

The receiving contract also reserves a separate deformation/vertex payload. That deformation complex, its normalized basis, and its Euler–Lagrange map are neither defined nor used in this paper. Their rank and topological statements are therefore deferred to the residual-mixing paper rather than introduced as a second theorem here. In particular, no deformation class is relabelled as a third dynamical particle branch.

The obstruction neither invalidates Theorem 4.1 nor computes a branch mixing table. It is scoped to the declared physical principal anchor on the exact contractible rank-46 graph carrier. It does not rule out a different leading cohomology carrier, a noncontractible or mixed-bundle curvature mapping cylinder, or a separately tagged nonlocal reduced-mode splitting. The next exact choice is therefore to retain the unsplit complex or construct the smallest noncontractible filtered enlargement before lifting the nonlinear brackets.

8 Bounded cyclic deformation status

The constant-field and first-jet physical-action primitives do not decide the full BV deformation problem: cyclic completion couples fields, ghosts and antifields, and a choice of lower primitive may hide kernel directions needed on the next filtration page. We therefore use the summed pre-reduction input PBW order and solve the zero and first full-BV pages simultaneously. The allowed coordinate changes are degree-zero, derivative-aware cyclic super-cotangent Taylor maps (F_2, F_3) of nonnegative filtered order.

Let A_0 denote the complete zero-page coboundary map, let B_a be the order-one contribution of a zero-page map on derivative axis a , and let C_a be the associated-graded first-jet coboundary. If r_a is the first-page residual after one certified zero-page primitive, changing that primitive by x and adding first-jet maps y_a requires

$$A_0 x = 0, \quad B_a x + C_a y_a = r_a, \quad a = 0, 1, 2, 3. \quad (49)$$

The first equation retains the complete kernel of A_0 ; freezing the earlier 67-term primitive would give a false single-coordinate obstruction.

Proposition 8.1 (Second-jet disposition of the former obstruction). *Let $\mathfrak{w}$ be the stored normalized 22-row functional. Its exhaustive transpose replay remains exact on the tested zero/first-jet subcomplex — 5,984 zero-page and 14,998 first-jet columns per axis — on which it is unaffected by the disposition below:*

$$\mathfrak{w}(A_0) = 0, \quad \mathfrak{w}(B_a, C_a) = 0 \quad (a = 0, 1, 2, 3), \quad \mathfrak{w}(r) = 1. \quad (50)$$

However, in the declared derivative-aware super-cotangent complex the admissible base column

$$F_2^4(D_1 D_1 \phi_{27}, \phi_{28}) \quad (51)$$

has five cyclic cotangent components. Native coderivation replay produces 252 page-one terms and

$$\mathfrak{w}(\delta F_2) = \frac{755}{9}. \quad (52)$$

Thus $\mathfrak{w}$ does not descend to the cokernel dual of the complete summed-input-order-two cyclic complex, and the earlier order-two nonremovability conclusion is withdrawn.

The mechanism is a lower-order commutator tail created by formal-adjoint cotangent transposition followed by Berger PBW reduction. Consequently the pre-reduction input-order filtration does not prevent a second-input-jet column from entering output page one. Multiplication of the displayed column by $-9/755$ cancels the old normalized witness value, but this is not a complete primitive: a direct lift of the physical order-two solution together with the three known ghost shears still leaves 1,380 exact page-zero full-BV residual coefficients. The complete bounded cyclic class is therefore *open*, not proved trivial and not proved nonzero. Opening q_4 would not answer this unresolved order-two coboundary problem.

9 Interpretation and nonclaims

At representative level, the retained classical tensor contains both a gravity-equation response to one gravity and two Maxwell inputs and a Maxwell-equation response to two gravity and one Maxwell input. After lowering, both are g, g, A, A quartic coefficients. They remain visible under the frozen exact cyclic transfer and satisfy the ternary homotopy identity. If a noncontractible enlargement receives an admissible branch projector and nonlinear lift—or an explicitly declared unsplit observable replaces the branch table—Equation (39) will decide whether this representative closes on the Einstein-like branch, couples it to the extra-Weyl branch, or vanishes on residual cohomology.

Several stronger conclusions do not follow.

1. The bracket changes no unary kinetic operator. It therefore introduces no new negative kinetic direction at this order, but this is not a unitarity or positive-Hilbert-space theorem.
2. A nonzero retained bracket is not yet a photon or graviton scattering amplitude. Residual projection, causal state selection, and asymptotic or relational observables are still required.
3. Nonvanishing of this coefficient representative alone proves neither a nonzero operation on ℓ_1 -cohomology nor a class in the unrestricted cyclic deformation complex. The complete constant-field page is removable by the exact F_3 primitive of Proposition 4.2, and the first-jet

physical page is removable by Proposition 4.3. The exact physical primitive extends through input order two, but Proposition 8.1 shows that the former full-BV obstruction witness is invalid on an admissible second-jet column. The complete bounded cyclic class remains open.

4. Deformation/vertex classes are not one-particle states and are not included in the dynamical branch list used here.
5. The vanishing one-particle cohomology of the separate free conformal cylinder problem is not converted here into an interacting vacuum theorem.
6. The observer apparatus imports the repaired q_2 only as a probe-limit baseline. Its extended apparatus q_1 , cyclicity, retarded Green operator, and interacting observer morphism are not certified; the representation- complete common-action carrier is separately obstructed. No relational promotion follows from the retained bracket.
7. No loop coefficient, anomaly cancellation, QME restoration, Hadamard state, renormalized product, or quantum interaction has been constructed.
8. No arity-four or convergent all-orders L_∞ or Cartan theorem is claimed.

The appropriate provisional interpretation is therefore a retained deformation/vertex representative rather than a particle interaction. The mixing table and deformation-cohomology calculation are the two distinct next tests.

10 Reproducibility and certificate map

The algebraic theorem is pinned to the classical hardening commit `e99d0c1d39490de5261fc6ca1dc2aeaa0d149655`. Its principal artifacts are

Artifact	Role
BERGER_COUPLED_K_CARTAN_THROUGH_ARITY_THREE	full/retained cyclic causal K -Cartan theorem
BERGER_RETAINED_MIXED_ELL3_INDEPENDENT_ACCEPTANCE	exact independent $64 \rightarrow 36$ transfer and mutation witnesses
BERGER_RETAINED_MIXED_ELL3_PHYSICAL_CYCLICITY	independent degree-zero lowered cyclicity and pairing-weight mutation
BERGER_RETAINED_MIXED_ELL3_FULL_BV_CYCLICITY	independent ghost/antifield completion and full retained BV cyclicity
BERGER_RETAINED_MIXED_ELL3_CONSTANT_FIELDED_REDEFINITION_V1	exact constant-field physical-action cyclic-redefinition primitive
BERGER_RETAINED_MIXED_ELL3_FIRST_JET_REDEFINITION_V1	exact coupled first-jet physical-action primitive
BERGER_RETAINED_MIXED_ELL3_ZERO_JET_GHOST_SHEAR_COMPLETION_V1	exact zero-PBW full-BV primitive after the certified Maxwell ghost shears
BERGER_RETAINED_MIXED_ELL3_SECOND_JET_EXACT_PRIMITIVE_V1	exact degree-zero physical-action primitive through input order two
BERGER_RETAINED_MIXED_ELL3_POSITIVE_JET_FULL_BV_OBSTRUCTION_V1	historical exact zero/first-jet screen; order-two interpretation superseded
BERGER_RETAINED_MIXED_ELL3_SECOND_JET_WITNESS_DISPOSITION_V1	exact $755/9$ second-jet invalidation; complete bounded cyclic class open
BERGER_FILTERED_CYCLIC_BRANCH_EXTENSION_OBSTRUCTION_V1	invariant unary branch-splitting obstruction and minimal page repair
NONLINEAR_SOURCE_TRANSFER_TANGENT_CONE_DICTIONARY_V1	$D^2E-q_2-\ell_2-\ell_3$ normalization and obstruction naturality
BERGER_RESIDUAL_MIXED_ELL3_BRANCH_PROJECTION_READINESS_V2	historical fail-closed input contract for the mixing table
BERGER_RETAINED_36_RESIDUAL_BRANCH_LOCAL_PROJECTOR_OBSTRUCTION_V1	normalized obstruction to the canonical local 36-row projector
BERGER_RETAINED_46_STF2_PROLONGATION_BRANCH_CARRIER_V1	exact local cyclic graph carrier; physical-anchor projector tested below
BERGER_RETAINED_46_STF2_PRINCIPAL_BRANCH_ANCHOR_V1	normalized principal nonsplitting witness
BERGER_RETAINED_46_STF2_PHYSICAL_HELICITY_FILTERED_QUOTIENT_V1	exact rank-two projective helicity module inside the six-class null quotient
BERGER_RETAINED_46_STF2_SUBPRINCIPAL_BRANCH_ANCHOR_OR_OBSTRUCTION_V1	normalized V_2 filtered-lift obstruction for the plus polarization

The scoped reproduction commands are

```

PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  berger_coupled_k_cartan_through_arity_three.py --check --guards
PYTHONPATH=. python3 \
  d_quotient_classical/backreacted_clock/\
  verify_berger_coupled_k_cartan_through_arity_three.py
PYTHONPATH=quantum-weyl python3 -m \
  transfer.berger_retained_mixed_ell3_acceptance_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
  transfer.verify_berger_retained_mixed_ell3_acceptance

```

```

PYTHONPATH=quantum-weyl python3 -m \
    transfer.berger_retained_mixed_ell3_physical_cyclicity_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_berger_retained_mixed_ell3_physical_cyclicity
PYTHONPATH=quantum-weyl python3 -m \
    transfer.berger_retained_mixed_ell3_ghost_antifield_cyclicity_certificate --check
PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_berger_retained_mixed_ell3_ghost_antifield_cyclicity
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    berger_retained_mixed_ell3_constant_field_redefinition.py --check
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_constant_field_redefinition.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    berger_retained_mixed_ell3_first_jet_redefinition.py --check
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_first_jet_redefinition.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_zero_jet_ghost_shear_completion.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_second_jet_exact_primitive.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_positive_jet_full_bv_obstruction.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_positive_jet_full_bv_obstruction_exhaustive.py \
    --workers 8
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_mixed_ell3_second_jet_witness_disposition.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_filtered_cyclic_branch_extension_obstruction.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_nonlinear_source_transfer_tangent_cone_dictionary.py
PYTHONPATH=quantum-weyl python3 -m \
    transfer.berger_residual_ell3_branch_projection_readiness_v2 --check
PYTHONPATH=quantum-weyl python3 -m \
    transfer.verify_berger_residual_ell3_branch_projection_readiness_v2
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_36_residual_branch_local_projector_obstruction.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    berger_retained_46_stf2_prolongation_branch_carrier.py --check --guards
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    verify_berger_retained_46_stf2_prolongation_branch_carrier.py
PYTHONPATH=. python3 \
    d_quotient_classical/backreacted_clock/\
    berger_retained_46_stf2_principal_branch_anchor.py --check --guards
PYTHONPATH=. python3 \

```



```

d_quotient_classical/backreacted_clock/\
verify_berger_retained_46_stf2_principal_branch_anchor.py
PYTHONPATH=. python3 \
d_quotient_classical/backreacted_clock/\
berger_retained_46_stf2_physical_helicity_filtered_quotient.py \
--check --guards
PYTHONPATH=. python3 \
d_quotient_classical/backreacted_clock/\
verify_berger_retained_46_stf2_physical_helicity_filtered_quotient.py
PYTHONPATH=. python3 \
d_quotient_classical/backreacted_clock/\
berger_retained_46_stf2_subprincipal_branch_anchor_or_obstruction.py \
--check --guards
PYTHONPATH=. python3 \
d_quotient_classical/backreacted_clock/\
verify_berger_retained_46_stf2_subprincipal_branch_anchor_or_obstruction.py

```

The companion machine-readable claim map records the dependency hashes, positive claims, and explicit nonclaims. The scoped frozen-SDR representative and cyclicity statements remain theorem-frozen. The former filtered-cyclic nonremovability statement is superseded; the complete bounded cyclic class is explicitly open.

11 Outlook

Paper 11 now has two mathematical continuations. The first is to decide between the unsplit retained complex and a noncontractible filtered or mixed-bundle enlargement, then materialize the relevant cyclic q_2, q_3 lift and compute the corresponding version of Equation (39). A nonzero Einstein/extra-Weyl entry would turn the additional Weyl physics into an explicit interaction channel; a block-diagonal table would establish nonlinear closure of the Einstein-like direction through this order; and a central topological action would isolate the vertex theory from the dynamical carrier. The second is to test whether the selected mixed representative defines a nonzero operation on ℓ_1 -cohomology or a class in an unrestricted cyclic deformation complex. Its physical action through input order two is exact, while the old full-BV dual screen is invalidated by Proposition 8.1. The next deformation calculation is therefore the coupled full-BV zero/first/second-jet coboundary-cokernel problem, not an enlargement to q_4 .

The algebraic theorem does not need to wait for that verdict. What the mixing table changes is its physical interpretation, not the existence of the nonzero mixed representative for the frozen cyclic SDR.

A Exact inclusion columns for the displayed witnesses

In Table 2, $\alpha = (\alpha_0, \alpha_1, \alpha_2, \alpha_3)$ denotes the ordered PBW derivative word in the four-generator order of the carrier certificate. The table is generated deterministically from the complete 64×36 inclusion record; it contains 106 exact operator coefficients. The five columns shown are the three dressed gravity inputs used by Equations (20)–(21), the gravity-equation output, and its dressed pairing partner.

Table 2: Exact full-carrier inclusion columns for every dressed gravity coordinate used by the two headline witnesses or their pairing partners. Each row denotes the term $c\partial^\alpha e_{\text{full}}$ in $\iota(e_{\text{retained}})$.

Retained ID	Row	Full-carrier ID	α	Exact c
h_hat_13	9	h_hat_13	(0, 0, 0, 0)	1
		bar_c_star_diff_1	(2, 0, 0, 1)	-1
		bar_c_star_diff_1	(0, 2, 0, 1)	1/3
		bar_c_star_diff_1	(0, 1, 1, 0)	-11*sqrt(10)/45
		bar_c_star_diff_1	(0, 0, 2, 1)	1
		bar_c_star_diff_1	(0, 0, 0, 1)	-475/48
		bar_c_star_diff_1	(0, 0, 0, 3)	1
		bar_c_star_diff_2	(0, 0, 0, 0)	-79391*sqrt(10)/43200
		bar_c_star_diff_2	(2, 0, 0, 0)	-31*sqrt(10)/60
		bar_c_star_diff_2	(0, 2, 0, 0)	11*sqrt(10)/30
		bar_c_star_diff_2	(0, 1, 1, 1)	-2/3
		bar_c_star_diff_2	(0, 0, 2, 0)	11*sqrt(10)/90
		bar_c_star_diff_2	(0, 0, 0, 2)	9*sqrt(10)/5
		bar_c_star_diff_3	(2, 1, 0, 0)	-1
		bar_c_star_diff_3	(0, 1, 0, 0)	-329/2160
		bar_c_star_diff_3	(0, 3, 0, 0)	1
		bar_c_star_diff_3	(0, 1, 2, 0)	1
		bar_c_star_diff_3	(0, 1, 0, 2)	1/3
		bar_c_star_diff_3	(0, 0, 1, 1)	2*sqrt(10)/45
		bar_c_star_tau	(1, 1, 0, 1)	-2/3
		bar_c_star_tau	(1, 0, 1, 0)	-71*sqrt(10)/180
		bar_c_star_sigma	(2, 1, 0, 1)	1/3
		bar_c_star_sigma	(2, 0, 1, 0)	71*sqrt(10)/360
		bar_c_star_sigma	(0, 3, 0, 1)	-1/3
		bar_c_star_sigma	(0, 2, 1, 0)	-71*sqrt(10)/360
		bar_c_star_sigma	(0, 1, 2, 1)	-1/3
		bar_c_star_sigma	(0, 1, 0, 1)	34/9
		bar_c_star_sigma	(0, 1, 0, 3)	-1/3
		bar_c_star_sigma	(0, 0, 1, 0)	71*sqrt(10)/81
		bar_c_star_sigma	(0, 0, 3, 0)	-71*sqrt(10)/360
		bar_c_star_sigma	(0, 0, 1, 2)	-13*sqrt(10)/24
h_hat_01	4	h_hat_01	(0, 0, 0, 0)	1
		bar_c_star_diff_1	(1, 0, 0, 0)	2561/720
		bar_c_star_diff_1	(3, 0, 0, 0)	1
		bar_c_star_diff_1	(1, 2, 0, 0)	-1/3
		bar_c_star_diff_1	(1, 0, 2, 0)	-1
		bar_c_star_diff_1	(1, 0, 0, 2)	-1
		bar_c_star_diff_2	(1, 1, 1, 0)	2/3
		bar_c_star_diff_2	(1, 0, 0, 1)	-77*sqrt(10)/60
		bar_c_star_diff_3	(1, 1, 0, 1)	2/3
		bar_c_star_diff_3	(1, 0, 1, 0)	107*sqrt(10)/180
		bar_c_star_tau	(2, 1, 0, 0)	-1/3
		bar_c_star_tau	(0, 1, 0, 0)	-31/9
		bar_c_star_tau	(0, 3, 0, 0)	1
		bar_c_star_tau	(0, 1, 2, 0)	1
		bar_c_star_tau	(0, 1, 0, 2)	1
		bar_c_star_tau	(0, 0, 1, 1)	31*sqrt(10)/30
		bar_c_star_sigma	(3, 1, 0, 0)	-1/3
		bar_c_star_sigma	(1, 1, 0, 0)	-31/27
		bar_c_star_sigma	(1, 3, 0, 0)	1/3
		bar_c_star_sigma	(1, 1, 2, 0)	1/3
h_hat_03	6	h_hat_03	(0, 0, 0, 0)	1
		bar_c_star_diff_1	(1, 1, 0, 1)	2/3
		bar_c_star_diff_1	(1, 0, 1, 0)	-3*sqrt(10)/20
		bar_c_star_diff_2	(1, 1, 0, 0)	3*sqrt(10)/20
		bar_c_star_diff_2	(1, 0, 1, 1)	2/3
		bar_c_star_diff_3	(1, 0, 0, 0)	9/80
		bar_c_star_diff_3	(3, 0, 0, 0)	1

Continued on next page

Table 2 (continued)

Retained ID	Row	Full-carrier ID	α	Exact c
		bar_c_star_diff_3	(1, 2, 0, 0)	-1
		bar_c_star_diff_3	(1, 0, 2, 0)	-1
		bar_c_star_diff_3	(1, 0, 0, 2)	-1/3
		bar_c_star_tau	(2, 0, 0, 1)	-1/3
		bar_c_star_tau	(0, 2, 0, 1)	1
		bar_c_star_tau	(0, 0, 2, 1)	1
		bar_c_star_tau	(0, 0, 0, 3)	1
		bar_c_star_sigma	(3, 0, 0, 1)	-1/3
		bar_c_star_sigma	(1, 2, 0, 1)	1/3
		bar_c_star_sigma	(1, 0, 2, 1)	1/3
		bar_c_star_sigma	(1, 0, 0, 3)	1/3
h_hat_star_00	13	h_hat_star_00	(0, 0, 0, 0)	1
h_hat_00	3	h_hat_00	(0, 0, 0, 0)	1
		bar_c_star_diff_1	(2, 1, 0, 0)	-1/2
		bar_c_star_diff_1	(0, 1, 0, 0)	71/480
		bar_c_star_diff_1	(0, 3, 0, 0)	1/6
		bar_c_star_diff_1	(0, 1, 2, 0)	1/6
		bar_c_star_diff_1	(0, 1, 0, 2)	1/6
		bar_c_star_diff_2	(2, 0, 1, 0)	-1/2
		bar_c_star_diff_2	(0, 2, 1, 0)	1/6
		bar_c_star_diff_2	(0, 1, 0, 1)	$-\sqrt{10}/20$
		bar_c_star_diff_2	(0, 0, 1, 0)	-3/160
		bar_c_star_diff_2	(0, 0, 3, 0)	1/6
		bar_c_star_diff_2	(0, 0, 1, 2)	1/6
		bar_c_star_diff_3	(2, 0, 0, 1)	-1/2
		bar_c_star_diff_3	(0, 2, 0, 1)	1/6
		bar_c_star_diff_3	(0, 0, 2, 1)	1/6
		bar_c_star_diff_3	(0, 0, 0, 1)	3/160
		bar_c_star_diff_3	(0, 0, 0, 3)	1/6
		bar_c_star_tau	(3, 0, 0, 0)	1/2
		bar_c_star_tau	(1, 2, 0, 0)	-5/6
		bar_c_star_tau	(1, 0, 2, 0)	-5/6
		bar_c_star_tau	(1, 0, 0, 2)	-5/6
		bar_c_star_sigma	(2, 2, 0, 0)	1/6
		bar_c_star_sigma	(2, 0, 2, 0)	1/6
		bar_c_star_sigma	(2, 0, 0, 2)	1/6
		bar_c_star_sigma	(0, 2, 0, 0)	-1/3
		bar_c_star_sigma	(0, 4, 0, 0)	-1/6
		bar_c_star_sigma	(0, 2, 2, 0)	-1/3
		bar_c_star_sigma	(0, 2, 0, 2)	-1/3
		bar_c_star_sigma	(0, 1, 1, 1)	$\sqrt{10}/10$
		bar_c_star_sigma	(0, 0, 2, 0)	1/3
		bar_c_star_sigma	(0, 0, 4, 0)	-1/6
		bar_c_star_sigma	(0, 0, 2, 2)	-1/3
		bar_c_star_sigma	(0, 0, 0, 2)	-3/40
		bar_c_star_sigma	(0, 0, 0, 4)	-1/6

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